

A Transparent Spatiotemporal Framework for Modeling Burn-State Transitions

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1 Problem Overview

The **objective** of this project is to model daily burn-state transitions for each pixel in a raster landscape. Rather than only predicting the burn state of a pixel at the next time step, this project focuses on predicting the transition from the current burn state to the next-day burn state. This transition-based formulation provides greater transparency because it directly represents how pixels move among different burn states and allows the model to identify the environmental, weather, and neighborhood conditions associated with each type of state change.

Each pixel can take one of five possible burn states:

- OU: outside unburned,
- IH: intense heat,
- SH: scattered heat,
- OH: isolated heat,
- IU: interior unburned.

Let

$$Y_{i,t} \in \{OU, IH, SH, OH, IU\}$$

denote the burn state of pixel i on day t . The primary response variable is the one-day burn-state transition:

$$T_{i,t \rightarrow t+1} = Y_{i,t} \rightarrow Y_{i,t+1}.$$

For example, possible transitions include

$$OU \rightarrow IH, \quad OU \rightarrow SH, \quad IH \rightarrow IU, \quad SH \rightarrow IU, \quad IU \rightarrow IU.$$

This project does not focus on developing a black-box deep-learning model. Instead, it focuses on developing a **transparent descriptive and predictive statistical framework** to help explain how burn states transition over time and how these transitions are influenced by static landscape characteristics, dynamic weather conditions, and the burn-state histories of neighboring cells.

2 Transition-Based Modeling Framework

Let $N(i)$ denote the eight-cell neighborhood surrounding pixel i , and let k denote the number of lag days used as historical information. Let X_i represent static environmental covariates for pixel i , such as fuel type, topography, vegetation, slope, aspect, elevation, and other landscape variables. Let $W_{i,t:t-k}^{obs}$ represent observed weather or fire-weather conditions from day $t - k$ through day t , and let $W_{i,t+1}^{fct}$ represent forecasted weather or fire-weather conditions for the prediction day $t + 1$.

For one-day-ahead transition prediction, the model can be written as:

$$P\left(T_{i,t \rightarrow t+1} \mid Y_{i,t:t-k}, Y_{N(i),t:t-k}, X_i, W_{i,t:t-k}^{obs}, W_{i,t+1}^{fct}\right).$$

Equivalently,

$$P\left(Y_{i,t} \rightarrow Y_{i,t+1} \mid Y_{i,t:t-k}, Y_{N(i),t:t-k}, X_i, W_{i,t:t-k}^{obs}, W_{i,t+1}^{fct}\right).$$

This formulation makes the current burn state $Y_{i,t}$ explicit and models the probability that the pixel transitions into a new state on day $t + 1$. During model training, historical burn-state maps provide observed transitions from $Y_{i,t}$ to $Y_{i,t+1}$. During inference, the trained model uses the most recent observed burn states, recent observed weather, static environmental covariates, and forecasted weather for day $t + 1$ to forecast the transition of each pixel.

3 Step 1: Define the Problem as a Supervised Spatiotemporal Transition Problem

The problem is framed as a supervised spatiotemporal classification problem on a raster lattice. Each pixel is treated as a cell, each day is treated as a time step, and the response variable is the transition from the pixel's current burn state to its next-day burn state.

The one-day-ahead transition rule can be written conceptually as:

transition from day t to day $t + 1 = f(\text{recent focal-pixel burn-state history through day } t,$
 $\text{recent neighborhood burn-state history through day } t,$
 $\text{static environmental covariates,}$
 $\text{observed recent weather through day } t,$
 $\text{forecasted weather for day } t + 1).$

This formulation is similar to a probabilistic cellular automaton because each pixel's next state depends on its own state history and the state history of nearby cells. However, the transition rule is learned from data rather than manually specified, and it explicitly incorporates both static landscape variables and dynamic weather information.

4 Step 2: Define the Prediction Target as a State Transition

The prediction target is the transition from the current day to the next day:

$$Y_{i,t} \rightarrow Y_{i,t+1}.$$

This is preferable to predicting only $Y_{i,t+1}$ because it preserves the starting state of each pixel. For example, predicting that a pixel is IH on day $t+1$ is informative, but predicting that it changed from OU to IH provides more information about fire spread and ignition. Similarly, predicting $IH \rightarrow IU$ describes a transition from active heat to interior unburned, which has a different interpretation from $OU \rightarrow IU$ or $IU \rightarrow IU$.

The transition-based response can be represented as a multi-class categorical variable. Since there are five possible burn states, there are up to

$$5 \times 5 = 25$$

possible one-day transitions. Some transitions may be rare or physically unlikely, but they can still be retained initially and later grouped or filtered based on empirical frequency and interpretation.

5 Step 3: Represent the Local Spatiotemporal Window

For each pixel-day observation, a local space-time window is extracted. This window contains the focal pixel and its eight neighboring pixels over the previous k days. For example, a 3×3 neighborhood and a five-day lag window gives a local input window of size:

$$3 \times 3 \times 5.$$

The focal-pixel history is represented as

$$Y_{i,t:t-k},$$

and the neighborhood history is represented as

$$Y_{N(i),t:t-k}.$$

Burn states can be encoded as integer categories or one-hot vectors. A one-hot encoding is useful because it avoids imposing artificial numerical order on the five burn states:

$$OU = [1, 0, 0, 0, 0], \quad IH = [0, 1, 0, 0, 0], \quad SH = [0, 0, 1, 0, 0],$$

$$OH = [0, 0, 0, 1, 0], \quad IU = [0, 0, 0, 0, 1].$$

The transition target can also be encoded categorically. For example:

$$OU \rightarrow OU, \quad OU \rightarrow IH, \quad OU \rightarrow SH, \quad \dots, \quad IU \rightarrow IU.$$

6 Step 4: Organize Predictors into Transparent Feature Groups

The predictors can be organized into four interpretable groups.

Focal-Pixel Burn-State History

The first group describes the recent burn-state history of the focal pixel. These variables answer questions such as: What state is the pixel currently in? How long has it been in this state? Has it recently transitioned from another state?

Examples include:

- current focal-pixel state $Y_{i,t}$,
- focal-pixel states at previous lag days $Y_{i,t-1}, Y_{i,t-2}, \dots, Y_{i,t-k}$,
- number of days since the pixel first entered an active heat state,
- number of days since the pixel last changed state.

Neighborhood Burn-State History

The second group describes the recent burn-state history of the eight neighboring pixels. These variables represent local spatial dependence and potential fire spread from nearby cells.

Examples include:

- number of IH neighbors on day t ,
- number of SH neighbors on day t ,
- number of OH neighbors on day t ,
- fraction of neighbors in active heat states,
- whether any neighboring cell was in IH, SH, or OH during the previous k days,
- number of neighboring cells that transitioned into active heat states during the previous k days.

Static Environmental Covariates

The third group contains static or slowly changing environmental variables. These variables help explain why transitions may be more likely in some locations than others.

Examples include:

- fuel type,
- vegetation class,
- elevation,
- slope,
- aspect,
- canopy cover,
- topographic position,
- distance to roads, firelines, or water bodies.

Observed and Forecasted Weather Conditions

The fourth group contains dynamic weather or fire-weather variables. Historical observed weather is used to describe recent conditions, while forecasted weather is used for one-day-ahead inference.

Examples include:

- observed temperature over the previous k days,
- observed relative humidity over the previous k days,
- observed wind speed and wind direction through day t ,
- cumulative precipitation over the previous k days,
- forecasted wind speed for day $t + 1$,
- forecasted wind direction for day $t + 1$,
- forecasted relative humidity and temperature for day $t + 1$.

Wind direction may be especially important because a burning neighbor located upwind of a focal pixel may have a different influence than a burning neighbor located downwind.

7 Step 5: Begin with a Transition Matrix Baseline

A useful first baseline is an empirical transition matrix:

$$P(Y_{i,t+1} \mid Y_{i,t}).$$

This transition matrix estimates the probability of each next-day state conditional only on the current state.

Table 1: Example structure of a burn-state transition matrix.

Current State / Next State	OU	IH	SH	OH	IU
OU					
IH					
SH					
OH					
IU					

This baseline provides a simple and interpretable benchmark. It answers questions such as: How often does OU remain OU? How often does OU transition into IH? How quickly do IH, SH, or OH pixels transition into IU?

8 Step 6: Add Neighborhood Transition Information

After the simple transition matrix, the next step is to include neighborhood information:

$$P(Y_{i,t} \rightarrow Y_{i,t+1} \mid Y_{i,t}, Y_{N(i),t}).$$

This allows the transition probability of the focal pixel to depend on the current states of the surrounding cells. For example, the transition $OU \rightarrow IH$ may be more likely when one or more neighboring cells are already in IH, SH, or OH. Similarly, the transition $IH \rightarrow IU$ may depend on how long neighboring cells have been burning or whether the local fire front has already passed.

Neighborhood variables can be represented directly using the eight neighboring states or summarized into interpretable count-based features.

9 Step 7: Add Lagged Transition History

The model can then be extended to include recent state histories and recent transition histories:

$$P(Y_{i,t} \rightarrow Y_{i,t+1} \mid Y_{i,t:t-k}, Y_{N(i),t:t-k}).$$

This step is important because fire behavior often depends not only on the current configuration but also on recent progression. For example, a pixel with an IH neighbor yesterday may have a different transition probability than a pixel with an IH neighbor five days ago.

Useful lagged transition features include:

- whether the focal pixel changed state during the previous k days,
- number of neighboring cells that transitioned from OU to IH during the previous k days,
- number of neighboring cells that transitioned from active heat to IU,
- days since the nearest active heat transition,
- maximum number of active-heat neighbors during the previous k days.

10 Step 8: Add Fuels, Topography, and Weather

The full transition model incorporates focal-pixel history, neighborhood history, static environmental variables, observed weather, and forecasted weather:

$$P(Y_{i,t} \rightarrow Y_{i,t+1} \mid Y_{i,t:t-k}, Y_{N(i),t:t-k}, X_i, W_{i,t:t-k}^{obs}, W_{i,t+1}^{fct}).$$

This formulation supports both descriptive and predictive goals. Descriptively, the model can identify which conditions are associated with specific burn-state transitions. Predictively, the model can forecast the most likely next-day transition for each pixel using current burn-state maps and forecasted weather.

11 Step 9: Choose an Initial Interpretable Model Class

Because the goal is transparency, the initial model should be interpretable and suitable for tabular transition features. Candidate models include:

- multinomial logistic regression,
- regularized multinomial regression,
- random forest,

- gradient boosted trees,
- XGBoost, LightGBM, or CatBoost.

A gradient boosted tree model may be a strong initial choice because it can capture nonlinear relationships and interactions among burn-state history, neighborhood conditions, fuels, topography, and weather while still allowing feature importance and partial dependence analysis.

Deep-learning approaches such as CNNs, ConvLSTMs, or U-Net-style raster-to-raster models can be considered later as comparison models. However, they are not the primary focus of this project because the goal is not only prediction accuracy but also transparent explanation of burn-state transitions.

12 Step 10: Validate Using Spatial and Temporal Holdouts

Validation should be designed carefully because nearby pixels and adjacent days are strongly autocorrelated. Randomly splitting pixel-day observations can lead to overly optimistic estimates of model performance.

Better validation strategies include:

- holding out entire fires,
- holding out later time periods,
- holding out spatial blocks,
- holding out fire-year combinations.

Evaluation should focus not only on overall classification accuracy but also on transition-specific performance. Important metrics include:

- confusion matrix of predicted versus observed transitions,
- per-transition precision and recall,
- performance for rare but important transitions such as OU $\rightarrow$ IH,
- map-based comparison of observed and predicted transition patterns,
- transition probabilities summarized by fuel type, topography, and weather condition.

13 Recommended Starting Formulation

A recommended starting formulation is a pixel-day transition table. Each row represents one pixel on one day. The target is the observed transition from day t to day $t + 1$:

$$T_{i,t \rightarrow t+1} = Y_{i,t} \rightarrow Y_{i,t+1}.$$

The predictors include the current and recent burn states of the focal pixel, the current and recent burn states of neighboring pixels, static environmental covariates, observed weather through day t , and forecasted weather for day $t + 1$.

This transition-based formulation provides a transparent framework for understanding how burn states evolve across a landscape. It also supports operational prediction because, during inference, the model can forecast the next-day transition of every pixel using the most recent burn-state map, recent observed weather, static landscape data, and forecasted weather for the next day.

Table 2: Example pixel-day transition training table.

Pixel	Day	Focal History	Neighborhood History	Covariates	Target
i	t	$Y_{i,t:t-k}$	$Y_{N(i),t:t-k}$	X_i, W^{obs}, W^{fcst}	$Y_{i,t} \rightarrow Y_{i,t+1}$